

Enhancing SME Performance through BI: A Comprehensive System Design for Bili Hu Coffees LLP

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Section – 1: Introduction of Company

Bili Hu Coffees LLP is an Indian specialty coffee roaster with headquarters in Bengaluru that is part of a rapidly growing craft coffee market. The Managing Director at Bili Hu, Mr. Vidyun Hebbar, informed me that while the company has successfully implemented Shopify as a platform for the direct sale of specialty coffee to consumers, other key business operations continue to be managed using separate systems. Shopify is used to process retail orders. Wholesale and inventory management information remains stored in Microsoft Excel documents. All accounting operations occur separately in Tally ERP — a common pattern among SMEs who utilize file-based, isolated data structures (Sheng et al., 2022; Zhang & Tansuhaj, 2023).

Mr. Hebbar also informed me that the Bili Hu roasting machine includes IoT functionality which provides batch-level logs. However, the logs generated by the machine have not been analysed to identify meaningful trends or patterns in production operations. This lack of analysis is reflective of many of the same challenges that SMEs experience when they implement IoT and analytics due to limited Business Intelligence (BI) capabilities (Bag et al., 2021; Lee et al., 2020). Consequently, Bili Hu continues to duplicate data across multiple platforms, to produce inconsistent reports, and to delay decision making, all of which are common problems experienced by businesses with fragmented data systems (Ransbotham et al., 2021).

The fragmentation of data systems being utilized by Bili Hu restricts the company's ability to pursue additional goals such as improved forecasting, personalized marketing of products to customers, reduced waste, and entry into new geographic markets. For this reason, a structured Business Intelligence system will be required to provide the integrated, data-driven decision making environment necessary to support the future success of Bili Hu (Hazen et al., 2020; Popovic et al., 2021).

Section – 2: Problem Identification

When I interviewed Managing Director Vidyun Hebbar, many operational, tactical and strategic barriers to using analytics at Coffee Coven emerged as a barrier to making good use of analytics (Popović et al., 2021; Troisi et al., 2022). Hebbar stated that its main system—Shopify—is separate from its second main system Excel and then there are additional third-party systems such as Tally ERP and IoT roasting logs. All of these operate separately, so the common issue is that "our systems do not talk to each other." As a result, employees spend significant amounts of time manually reconciling numbers, and the business has no real-time view of inventory levels, especially when there are overlapping wholesale and online orders (Hazen et al., 2020; Sheng et al., 2022).

Demand forecasting is also a major issue for Hebbar. He stated that all demand planning is done in Excel, since the team is comfortable with the tool, however, the team's reliance on Excel results in poor forecasts, leading to over-roasting and failing to meet demand, resulting in wasted resources, inconsistent production and delayed fulfilment of online orders during peak sales periods.

Hebbar also mentioned that Shopify provides very limited analytical capabilities, providing the team with an inadequate understanding of consumer behaviour. This limits the teams' ability to identify high value consumers, repeat customers and distinguish between subscription-based and one-time purchases, therefore, marketing decisions are typically made based on intuition versus data-driven insights (Ransbotham et al., 2021; Pratama & Firmansyah, 2021).

Hebbar also identified two key strategic barriers to scaling Coffee Coven's subscription model and expanding geographically: maintaining consistency in product quality and having an inability to track the digital journey of the customer. Additionally, the lack of GA4 limits their ability to analyse marketing performance or measure the progression through the funnel, limiting strategic decision-making (Wamba et al., 2020).

Section – 3: Literature Review

The above literature review examines the various tactical, strategic and operational problems encountered by Bili Hu Coffees LLP using contemporary empirical research, business intelligence theory, and current analytical paradigms. The literature review also discusses how Business Intelligence components can develop analytical maturity and enhance the ability of MSMEs to make decisions based on data, including data warehousing, data integration, predictive modeling, and digital behaviour analytics.

3.1 Application of BI to SMEs

Business Intelligence is a socio-technical system that combines data, processes, technology and organizational culture to transform unrefined data into useful knowledge. In line with the IDC definition, Business Intelligence facilitates fact-based strategic and operational

decision making. Research has shown that Business Intelligence improves SME performance by providing better access to information and analytical capacity (Popović et al., 2021). However, SMEs are frequently hampered by the lack of an integrated data environment, their intuitive decision-making approach, and their low IT maturity. Therefore, to adopt Business Intelligence, SMEs require high levels of organizational readiness, governance, and employee education (Chen et al., 2021; Mikalef et al., 2021).

3.2 Data Warehousing, ETL and Integration

Data Warehouses (DWs) enable the consolidation of disparate datasets (e.g., those found in Shopify, Excel, Tally ERP, and IoT logs) into a single analytical environment. ETL pipelines standardize the data and prepare it for analysis. By automating ETL, the quality of the data is enhanced and the speed at which insights can be generated is increased; SMEs without a DW are therefore constrained to only descriptive, siloed reporting (Wu et al., 2022).

3.3 OLAP and Descriptive Analytics

OLAP enables multi-dimensional analysis via drill-down, roll-up, and slice-and-dice operations. Descriptive analytics summarize past performance (Pratama & Firmansyah, 2021), but do not optimize future performance. SMEs relying entirely on Shopify or Excel have no visibility into profitability, seasonality, and SKU performance (Boons & Lüdeke-Freund, 2020).

3.4 Predictive Analytics

Predictive models including regression and time series forecasting (ARIMA, SARIMA, Prophet) transform historical data into forward looking insights and support improved demand planning and reduced waste (Taylor & Letham, 2020; Makridakis et al., 2020).

3.5 Data Mining

Clustering, classification, and association rule mining provide insights into customer segmentation, churn prediction, and cross sell opportunities areas that are highly relevant to Bili Hu's retention and SKU optimization needs (Lamba & Dubey, 2021; Wu et al., 2022).

3.6 GA4 and Digital Behavior Analysis

GA4 provides event-driven behavioral insights and allows analysts to analyse funnels, engagement, and conversions all three of which are currently absent in Bili Hu's digital analytics (Zhang & Tansuhaj, 2023; Sheng et al., 2022).

3.7 RapidMiner & AutoML

AutoML in RapidMiner removes technical barriers and allows for the rapid deployment of Business Intelligence systems (Wu et al., 2022).

Integration

There were several key findings regarding the use of Business Intelligence in Small and Medium-Sized Enterprises. These included the necessity for integrated data environments, the development of analytical maturity, the need for interpretability, and the utilization of customer and web analytics all these factors will influence the Business Intelligence system design.

Section 4: The State of Business Intelligence & Analytical Practice

My conversation with Managing Director Vidyun Hebbar illustrated many issues affecting Coffee Coven's use of analytic tools for effective application (Popovic et al., 2021; Troisi et al., 2022). Hebbar stated that because their systems (Shopify, Excel, Tally ERP and IoT Roasting Logs) "don't communicate" with each other, Coffee Coven employees must manually match their systems so they can see what their current levels of inventory are particularly during periods of overlapping wholesale and on-line sales (Hazen et al., 2020; Sheng et al., 2022).

All forecasting at Coffee Coven is done in an Excel spreadsheet, a method Hebbar indicated produces inaccuracies regularly. The result is Coffee Coven regularly over-rotates, under-predicts the potential for increased demand, wastes product and must delay fulfilling customer orders.

Shopify does provide some basic analytics and as such, Coffee Coven has no way of identifying repeat customers or understanding why there may be behaviour differences between its subscription and one-time purchase customers. Consequently, all of Coffee Coven's marketing decisions are made based on intuition, rather than data (Pratama & Firmansyah, 2021; Ransbotham et al., 2021).

Hebbar indicates he would like to grow Coffee Coven's subscriptions and increase its geographical reach; however, he believes quality control issues, along with the absence of Google Analytics 4 (GA4), are limiting these goals, and restricting his ability to analyse and understand the complete digital journey of his customers and make informed marketing decisions (Wamba et al., 2020).

Section 5: Proposed Business Intelligence Solution Design and Solutions

These business intelligence (BI) principles are being applied to the operational, tactical, and strategic issues facing Bili Hu Coffees LLP with a focus on the constraints of small to medium sized enterprises (SMEs):

- Data Integration
- ETL Automation
- Warehouse
- OLAP Analysis
- Predictive Modeling

- Dashboard-based Insights

5.1 BI System Architecture

The BI system architecture will include one, trusted platform which enables all team members to share sales, inventory and production data. Hebbar made it clear that when there are small differences between the information stored in Shopify and Excel "it creates a lot of internal confusion." Therefore, the proposed architecture includes an Azure Synapse Data Warehouse. An integrated data warehouse is considered a foundational element of BI systems and has been identified as a key BI principle in current BI literature (Hazin et al., 2020; Popovic et al., 2021).

5.2 Integrated Data Sources

Shopify, Excel, Tally ERP, IoT roasting logs, GA4, and marketing datasets will be consolidated into a single integrated environment within the BI system. This will eliminate redundancy, improve the quality of the data and enable analytics based on multiple sources including demand forecasting, profitability analysis and behavioural segmentation (Sheng et al., 2022).

5.3 ETL Pipeline

Hebbar noted that there were many differences between the excel files used by different team members. This further emphasizes the need for automatic ETL processes which clean and normalize the data prior to loading. As indicated in previous literature, using ETL pipelines to create high-quality data can greatly enhance an SME's ability to use their data for analytics (Wu et al., 2022). Additionally, the use of Azure Data Factory to connect Shopify, Tally ERP and IoT logs will decrease the manual labour required and address Hebbar's operational issues.

5.4 Data Warehouse & OLAP Cube

When asked about the reports he would like most, Hebbar said he would like to be able to compare how each SKU performed over time and identify early product trends. OLAP cubes provide the exact type of multi-dimensional analysis which he is seeking (Pratama & Firmansyah, 2021). In addition, adding RFM scores and cluster labels to the customer dimension provides alignment with current research demonstrating that segmentation increases SME marketing effectiveness (Dubey & Lamba, 2021; Mikalef et al., 2021).

5.5 Predictive Analytics

One of Hebbar's primary concerns is forecasting. Hebbar acknowledged that "our planning is typically reactive" due to the uncertainty of future demand and the inability to plan, particularly during peak seasons. Techniques such as SARIMA, Prophet and regression have been demonstrated to greatly enhance the accuracy of forecasts (Taylor & Letham, 2020; Makridakis et al., 2020). These models provide a direct benefit to the company's desire for

improved production planning. Additionally, churn prediction and association rules provide a response to Hebbar's desire to improve subscription retention, consistent with SME analytics literature (Wu et al., 2022).

5.7 Dashboards (Power BI + GA4)

Power BI dashboards support decision-making through forecasting, inventory management, customer segmentation, SKU profitability and IoT quality monitoring. GA4 dashboards support funnel analysis and attribution modeling (Chen et al., 2021; Wamba et al., 2020).

Category of KPIs	Metrics	Definitions/Logic	SME Current Status	Target Post-BI (6-12 Months)	Expected Business Impacts
Efficiency	Accuracies in Production Forecast (MAPE)	Deviation Between Actual and Forecasted Quantity of Roasted Beans	Heuristic use of Spreadsheets and High Volatility	Down by 10-15% Use of SARIMA/Prophet	Less Over/Under-Roasting; Less Waste; Better Inventory Stability.
Control Quality (IoT)	Variance of Roast Profiles	Standard Deviation of Bean Temperature Curves vs. Ideal Curve	No monitoring; Manual	Down by 20-25% Using IoT Analytics.	More consistent flavour; less rejection; better brand image
Performance of Stock	Sell Through for SKU (%)	%Quantity Sold vs. Stocked during cycle	Unstructured; Manual Checks	Up by 5-10% via OLAP and Clustering	Better SKU Rationalization; Fewer Dead SKUs
Behaviour of Customers	Customer Life Value	Total Revenue per Customer Across Reorder Cycles	Unknown; Spreadsheet	Up by 8-12% Via Segmentation and Retention Models	Higher Repeat Purchases; Increased Loyalty
Acquisition of Customers	CAC (Cost per Acquisition)	Marketing Spend / New Conversion	Shopify/ads manual tracking	Down by 10-15% Via GA4 Funnel Insights	More Efficient Marketing Spend
Expansion of Subscription	Retention rates of Active Subscribers	% Active Subscribers after 3-6 Months	No tracking	Up by 5-8% with Churn Prediction	Stable Recurring Revenue
Reporting Efficiency	BI Report Generation Time	Time to Create Key Dashboards	Manual Spreadsheet Works	Down from 3-5 hours -> <30 minutes	Faster Decision Making; Improved Productivity
Growth Strategic	New Market Profitability of SKU	OP per SKU by region	No segmentation of Profitability	Uplift by 5-8% After OLAP Benchmarking	Better decision on expansion into new markets.

5.8 Decision Support Outcomes

Problem	BI Solution	Effect
Inaccurate forecasts	SARIMA + Prophet	Reduced waste; improved inventory accuracy
No segmentation	K-Means	Higher customer retention
Manual reporting	ETL automation	80% reduction in reporting time
Lack of acquisition analytics	GA4	Higher marketing ROI
Data silos	DW integration	Single source of truth
Poor SKU insight	OLAP cube	SKU rationalisation

5.9 Alignment with BI Theory

The system aligns with specifically DW architecture supports the BI lifecycle; OLAP supports multi-dimensional analysis; data mining supports segmentation, forecasting, churn prediction, etc.; GA4 supports behavioural insights; and RapidMiner AutoML supports SME friendly implementation. The components collectively form a theoretically based and practically feasible BI system (Lee et al., 2020; Bag et al., 2021).

Section 6: Implantation phase

Implementation of a Business Intelligence System at Bili Hu Coffees LLP will be conducted in a sequential manner as defined by the BI Life Cycle (Okoli & Wamba, 2021; Troisi et al., 2022). The phases are as follows:

- Phase 1: Define Key Performance Indicators (KPI's), BI objectives, define Data Sources and establish Governance.
- Phase 2: Develop Extract Transform Load (ETL) Pipelines using ADF to develop an Azure Synapse Star-Schema Warehouse.
- Phase 3: Build forecasting models, segmentation, churn prediction models, association rule mining and build OLAP Cubes.
- Phase 4: Deploy Power BI Dashboards, Google Analytics 4 (GA4) and provide BI Literacy Training for all stakeholders.

SECTION 6 — CONCLUSION AND BUSINESS INSIGHTS

During my conversation with Managing Director Vidyun Hebbar, I found out that while Bili Hu produces an incredible amount of useful data through many parts of their operation, most of this valuable information is currently not being used because of system fragmentation and limited access to analysis tools. Hebbar's biggest challenge is "to trust

the numbers and act upon them”, which is exactly what the proposed Business Intelligence (BI) System will allow him to do.

The BI System proposed will enable Bili Hu to transition from simple Descriptive Analytics to more advanced types of Analytical capabilities including integration of all data sources, automated Extract/Transform/Load (ETL), use of Predictive Models, and access to Google Analytics 4 (GA4) for Digital Behaviour Insights, which is consistent with the direction recommended by current BI Research (Mikalef et al., 2021; Wamba et al., 2020). The benefits of these new types of analytical capabilities at Bili Hu include more accurate Production Planning, consistent Roasting Quality, Improved Customer Retention, and more clearly defined Strategic Decision Making as the company continues to grow.

AI Acknowledgment

ChatGPT was used for idea generation, outline structuring, and refinement of section headings. All analysis, adaptation, and writing are the student’s own work. The use of AI complies with Brunel University’s guidance on the ethical and transparent use of generative AI in academic work (Brunel University London, 2025).

The following appendix provides verified company details to demonstrate the authenticity of this coursework.

Appendix : Company and Contact Details – Bili Hu Coffees LLP

Company Name: Bili Hu Coffees LLP

Registered Office Address: Bengaluru, Karnataka, India (as listed on MCA and company website)

Official Website: <https://bilihucoffee.com>

Public Business Contact: Email: hello@bilihucoffee.com

Managing Partner: Mr. Vidyun R. Hebbar – Managing Partner & Founder (as per MCA records) | Phone: +91 7906605943

Harvard Reference List

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